CORE PURE / COMPLEX NUMBERS

Loci and Regions in the Argand Diagram

Mark scheme 10 questions · 87 marks

Questions

Question	Marks by part	Marks	Page
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Question 3	(a)(i) 1 (a)(ii) 1 (b) 1 (c) 3 (d) 4	10	7
Question 4	(a)(i) 2 (a)(ii) 3 (b)(i) 3 (b)(ii) 3	11	9
Question 5	(a) 2 (b)(i) 2 (b)(ii) 2 (c) 5	11	11
Question 6	(a) 5 (b) 5 (c) 2	12	13
Question 7		7	15
Question 8	(a) 2 (b)(i) 2 (b)(ii) 4	8	16
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Question 10	(a) 2 (b) 1 (c) 2 (d) 2	7	19
	Total	**87**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes.

ft	follow through	**AG**	answer given in the question
cao	correct answer only, with no follow-through	**cso**	correct solution only: no errors in the part
oe	or an equivalent form	**awrt**	any answer that rounds to this value
soi	seen, or implied by later working	**isw**	ignore working after a correct answer
SC	special case: credit outside the main method	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

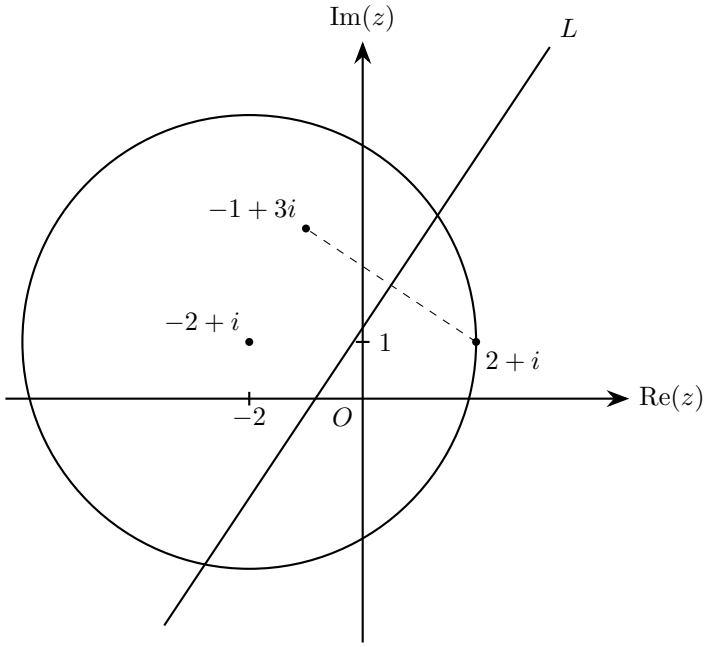
- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- **[*n*]** ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

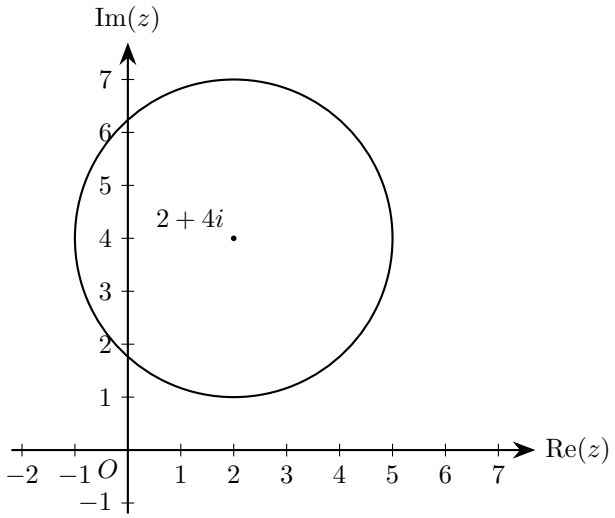
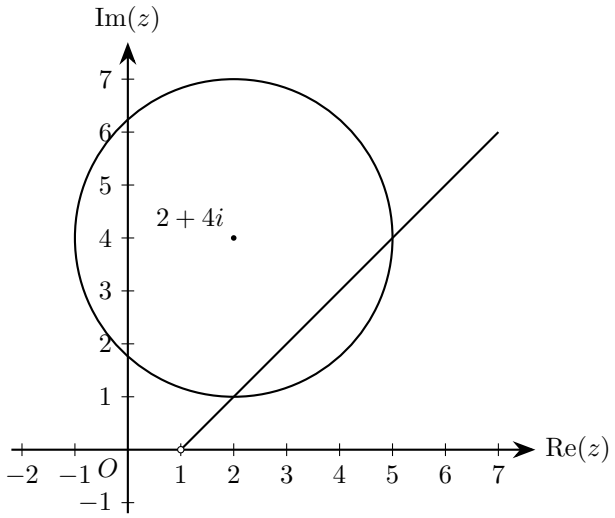
Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of g

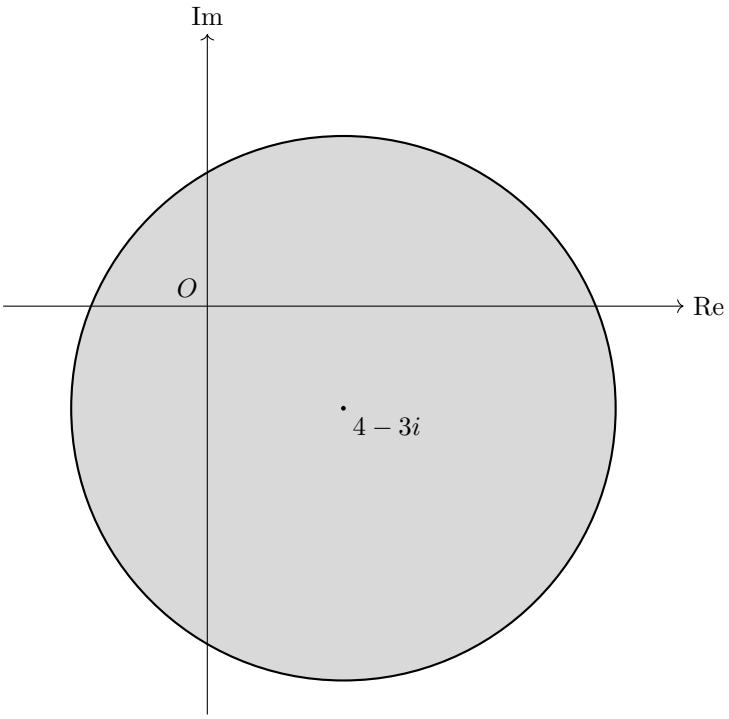
- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1(a)	$\{z \in \mathbb{C} : \lvert z+2-i\rvert = 4\}$	**B1 B1**	**B1** $\lvert z+2-i\rvert$ oe, such as $\lvert z-(-2+i)\rvert$. **B1** Equal to 4, inside set braces with z defined by a colon or bar. Condone $\in \mathbb{C}$ omitted.
		[2]	
1(b)			
	Shape	**B1**	Straight line, drawn as the perpendicular bisector of the segment from $2+i$ to $-1+3i$.
	Labelled point	**B1**	Through $\left(\frac{1}{2}, 2\right)$ with positive gradient, about $\frac{3}{2}$, crossing the Re axis between -2 and O; labelled L.
		[2]	
1(c)	$z = x+5i \Rightarrow z+1-2i = (x+1)+3i$ $\tan\frac{2\pi}{3} = \frac{3}{x+1} \Rightarrow x = \ldots$	**M1**	Uses $\mathrm{Im}(z) = 5$ and $\tan\frac{2\pi}{3}$ or $\tan\frac{\pi}{3}$ as the ratio of imaginary to real parts of $z+1-2i$, solving for x. Condone sign slips.
	$z = -1-\sqrt{3}+5i$	**A1**	oe, or awrt $-2.73+5i$, as a complex number.
		[2]	
	Total for Question 1	**6**	

Question	Scheme	Marks	Notes
2(a)			
	Shape	**B1**	A closed circle, freehand accepted.
	Centre and radius	**B1**	Centre at $2+4i$ and radius 3, shown by position against the scales or by labels.
		[2]	
2(b)			
	Half-line	**B1**	Half-line from 1 on the real axis at $\frac{\pi}{4}$, through about $(5,4)$, going up and right only; start shown excluded or not marked included.
		[1]	
2(c)	$(x-2)^2+(y-4)^2=9$, $y=x-1$ $\Rightarrow (x-2)^2+(x-5)^2=9 \Rightarrow x=\ldots$	**M1**	Substitutes the line into a circle equation of the form $(x-2)^2+(y-4)^2=9$, or eliminates x, and solves the quadratic by any means. Condone slips.

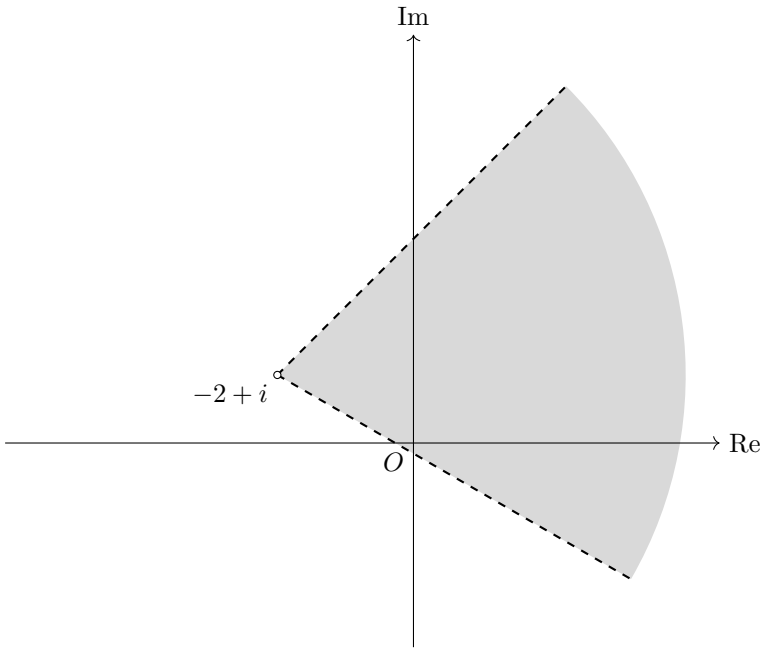
Question 2 continued

Question	Scheme	Marks	Notes
	$x = 5 \Rightarrow 5 + 4i$	**A1**	$5 + 4i$ or $(5, 4)$ taken as the furthest point; $(2, 1)$ need not be found.
	$\lvert w\rvert_{\max} = \sqrt{5^2 + 4^2} = \sqrt{41}$	**A1**	$\sqrt{41}$ or awrt 6.40.
		[3]	
	Total for Question 2	**6**	

Question	Scheme	Marks	Notes
3(a)(i)	$(4, -3)$	**B1**	Accept $4 - 3i$.
		[1]	
3(a)(ii)	8	**B1**	
		[1]	
3(b)			
	Shading	**B1**	Interior of a circle shaded, centre $4 - 3i$ in the fourth quadrant, enclosing O, with the centre or radius 8 indicated.
		[1]	
3(c)	$\lvert 4 - 3i \rvert = \sqrt{4^2 + 3^2} = 5$	**B1**	
	$\lvert z \rvert_{\max} =$ "5" $+ 8 = \ldots$	**M1**	Adds the radius to their distance from O to the centre, or maximises $\lvert z \rvert$ on the circle by another valid method.
	13	**A1**	
		[3]	
3(d)	$x + y = 9$, $(x-4)^2 + (y+3)^2 = 64$ $\Rightarrow x^2 - 16x + 48 = 0 \Rightarrow x = \ldots$	**M1**	Combines a line through $(3, 6)$ of gradient ± 1 with their circle equation and solves for the intersections by any means. Condone sign slips.
	$(4, 5)$, $(12, -3)$ $\Rightarrow \angle PCQ = \frac{\pi}{2}$	**A1**	Angle at the centre $\frac{\pi}{2}$ or $90°$, from correct intersection points.

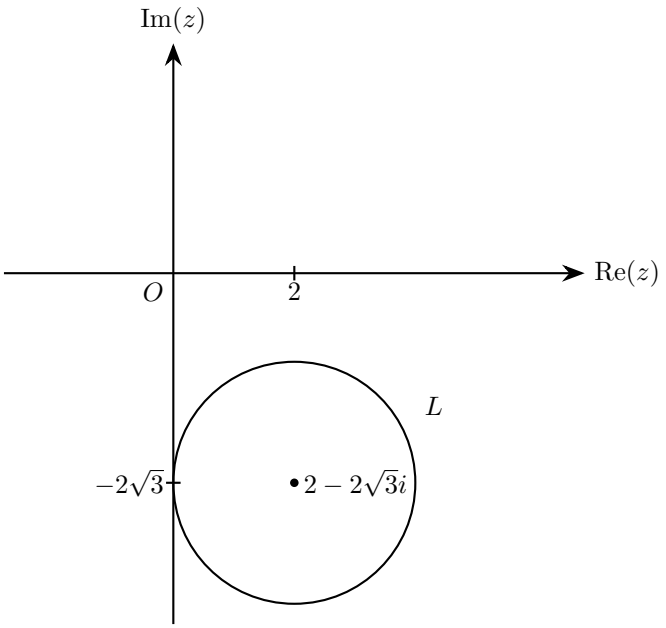
Question 3 continued

Question	Scheme	Marks	Notes
	Area $= \frac{1}{2} \times 8^2 \times$ "$\frac{\pi}{2}$" $\quad - \frac{1}{2} \times 8^2 \times \sin$ "$\frac{\pi}{2}$"	**dM1**	Uses sector minus triangle, $\frac{1}{2}r^2(\theta - \sin\theta)$, with $r = 8$ and their angle at the centre.
	$= 16\pi - 32 = 18.3$	**A1**	awrt 18.3.
		[4]	
	Total for Question 3	**10**	

Question	Scheme	Marks	Notes
4(a)(i)	$\|z-(-2+4i)\|=3$ $\|z+2-4i\|=3$	**B1 B1**	**B1** $\|z-(-2+4i)\|$ or $\|z+2-4i\|$ seen, with any right-hand side. **B1** Correct equation in any modulus form.
		[2]	
4(a)(ii)	$\|z+2-4i\|<3$	**B1**	With any inequality or equals sign.
	$\|z-(2+6i)\|<4$	**B1 B1**	**B1** With any inequality or equals sign. **B1** Strict $<$ in both, on inequalities for both circles.
		[3]	
4(b)(i)			
	Half-line	**B1**	Half-line from $-2+i$ at $\frac{\pi}{4}$ above the positive real direction; angle labelled or clearly in proportion.
	Half-line	**B1**	Half-line from $-2+i$ at $\frac{\pi}{6}$ below the positive real direction; angle labelled or clearly in proportion.
	Shading	**B1**	Region between the half-lines, to the right of $-2+i$, shaded or labelled. Condone solid boundaries and an unmarked vertex.
		[3]	

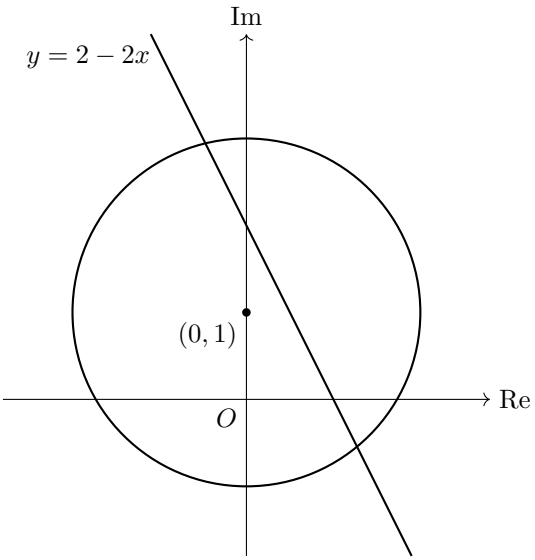
Question 4 continued

Question	Scheme	Marks	Notes
4(b)(ii)	$z+2-i=3+4i \Rightarrow \arg(3+4i)=\ldots$	**M1**	Substitutes $1+5i$ into $z+2-i$ and attempts its argument, or the gradient from $-2+i$ to $1+5i$.
	$\arctan\frac{4}{3} \approx 0.927$	**A1**	$\arctan\frac{4}{3}$, awrt 0.927 or 53.1°, or gradient $\frac{4}{3}$.
	$0.927 > \frac{\pi}{4} \approx 0.785$ $1+5i$ is not in the region	**A1ft**	ft their argument or gradient. Compares with $\frac{\pi}{4}$ (or $\tan\frac{\pi}{4}=1$), and with $-\frac{\pi}{6}$ where needed, and concludes correctly.
		[3]	
	Total for Question 4	**11**	

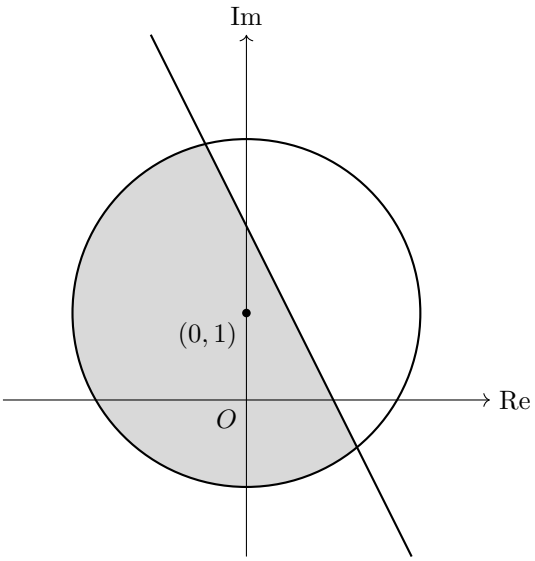
Question	Scheme	Marks	Notes
5(a)	$\lvert 2+2\sqrt{3}i\rvert = \sqrt{2^2+(2\sqrt{3})^2}$	**M1**	Squares the real and imaginary parts, adds and takes the square root. Condone one slip in squaring.
	$=4$	**A1**	
		[2]	
5(b)(i)			
	Shape	**B1**	A circle drawn wholly below the real axis.
	Centre and radius	**B1**	Centre $2-2\sqrt{3}i$ or $(2,-2\sqrt{3})$ indicated, and the circle touching the imaginary axis at $-2\sqrt{3}i$.
		[2]	
5(b)(ii)	$\sin\alpha = \frac{2}{4} \Rightarrow \alpha = \frac{\pi}{6}$ $\arg w = -\frac{\pi}{3} + \alpha$	**M1**	Uses the tangent from O to L, adding the angle between OC and the tangent to their arg of the centre, or finds the tangent's gradient by any method.
	$\arg w = -\frac{\pi}{6}$	**A1**	In terms of π.
		[2]	
5(c)	$\lvert 2+2\sqrt{3}i\rvert = 4$, $\arg = \arctan\sqrt{3} = \ldots$	**M1**	Finds the modulus and attempts an argument by trigonometry on the real and imaginary parts. The modulus may come from (a).
	$z^4 = 4e^{i\pi/3}$	**A1**	oe, such as $4\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$.

Question 5 continued

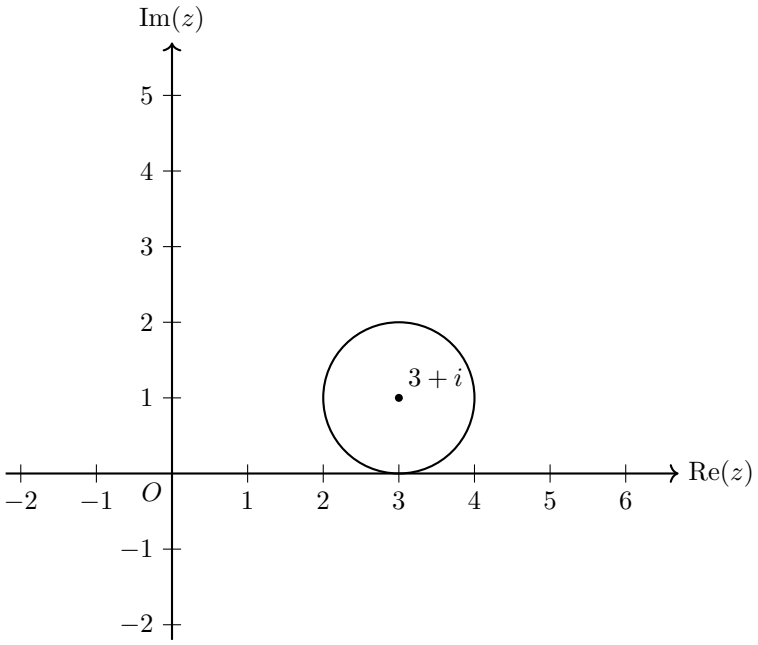
Question	Scheme	Marks	Notes
	$r = \text{“4”}^{1/4}$, $\theta = \dfrac{\text{“}\frac{\pi}{3}\text{”} + 2k\pi}{4}$	**M1**	Takes the fourth root of their modulus, and divides their argument plus $2k\pi$ by 4 for at least two values of k, or adds multiples of $\frac{\pi}{2}$ to one root's argument.
	$z = \sqrt{2}\,e^{i\pi/12},\ \sqrt{2}\,e^{7\pi i/12}$	**A1**	Any two correct roots; arguments need not be in range. Modulus $\sqrt{2}$ oe or awrt 1.41, arguments exact or awrt.
	$z = \sqrt{2}\,e^{-11\pi i/12},\ \sqrt{2}\,e^{-5\pi i/12}$	**A1**	All four roots in the form $re^{i\theta}$ with $-\pi < \theta \leq \pi$, and no others.
		[5]	
	Total for Question 5	**11**	

Question	Scheme	Marks	Notes
6(a)	$(x-1)^2+(y-1)^2$ $= \frac{1}{4}\left((x-4)^2+(y-1)^2\right)$	**M1 A1**	**M1** Writes both moduli in terms of x and y and squares both sides. Condone sign slips and the $\frac{1}{2}$ left unsquared. **A1** Any correct equation without square roots.
	$3x^2+3y^2-6y-9=0$ $\Rightarrow x^2+(y-1)^2=\ldots$	**dM1**	Expands and collects to $x^2+y^2+ax+by+c=0$ or a multiple, then completes the square or uses centre $\left(-\frac{a}{2},-\frac{b}{2}\right)$. Condone arithmetic slips.
	$x^2+(y-1)^2=4$, a circle	**A1**	Or $x^2+y^2-2y-3=0$, with the locus stated to be a circle.
	Centre $(0,1)$, radius 2	**A1**	Both, cso.
		[5]	
6(b)			
	Shape	**B1**	A circle, crossing both axes twice.
	Centre and radius	**B1**	Centre $(0,1)$ marked, and radius 2 shown or the circle through $3i$ and $-i$.
	$(x-2)^2+(y-3)^2=(x+2)^2+(y-1)^2$ $\Rightarrow -8x-4y+8=0$	**M1**	Squares the moduli in terms of x and y and simplifies to a linear equation, or finds the perpendicular bisector of $(2,3)$ and $(-2,1)$. Condone sign slips.
	$y=2-2x$	**A1**	oe, such as $2x+y=2$.

Question 6 continued

Question	Scheme	Marks	Notes
	Line $y = 2 - 2x$ on the same diagram	**B1**	Straight line of negative gradient through $(1, 0)$ and $(0, 2)$, cutting the circle twice. Intercepts need not be labelled.
		[5]	
6(c)			
	Shading	**B1**	Shaded region lies inside the circle.
	Shading	**B1**	Shading on the side of the line containing O, with only the intersection of the two regions shaded.
		[2]	
	Total for Question 6	**12**	

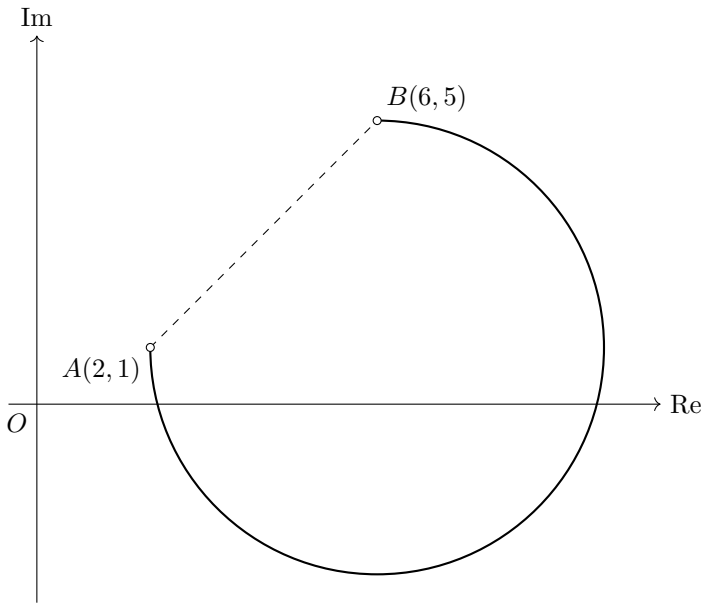
Question	Scheme	Marks	Notes
7	$(x-2)^2+(y+1)^2=9$	**B1**	oe, such as $x^2+y^2-4x+2y-4=0$.
	$(x+1)^2+(y-11)^2=9\left((x+1)^2+(y-3)^2\right)$	**M1**	Writes both moduli in terms of x and y and squares both sides. Condone sign slips.
	$(x+1)^2+(y-2)^2=9$	**A1**	oe, such as $x^2+y^2+2x-4y-4=0$.
	$(x-2)^2+(y+1)^2=(x+1)^2+(y-2)^2$ $\Rightarrow -6x+6y=0$	**M1**	Subtracts their circle equations, each with equal coefficients of x^2 and y^2, to reach a linear equation in x and y. Condone sign slips.
	$y=x$	**A1**	
	$(x-2)^2+(x+1)^2=9$ $\Rightarrow x^2-x-2=0\Rightarrow x=\ldots$	**dM1**	Substitutes their linear equation into either circle and solves the three-term quadratic by any means, leading to $x=\ldots$ or $y=\ldots$.
	$(2,2)$ and $(-1,-1)$	**A1**	Both points and no others.
	Total for Question 7	**7**	

Question	Scheme	Marks	Notes
8(a)			
	Shape	**B1**	A circle, roughly round.
	Centre and radius	**B1**	Centre at $3+i$ and radius 1, shown by position against the scales or by labels.
		[2]	
8(b)(i)	$(x-3)^2+(mx-1)^2=1$ $\Rightarrow (1+m^2)x^2-(6+2m)x+9=0$ $(6+2m)^2-36(1+m^2)=0 \Rightarrow m=\ldots$	**M1**	Uses a tangency condition, such as a zero discriminant for their quadratic in x or the distance from $(3,1)$ to $y=mx$ equal to 1, and solves for m.
	$m=\frac{3}{4}$ ($m=0$ rejected) $\alpha=\arctan\frac{3}{4}=0.644$	**A1**	$\arctan\frac{3}{4}$, awrt 0.644 or 36.9°. A0 if $\alpha=0$ is also given as an answer.
		[2]	
8(b)(ii)	$(x-3)^2+\left(\text{“}\frac{3}{4}\text{”}x-1\right)^2=1$ $\Rightarrow 25x^2-120x+144=0 \Rightarrow x=\ldots$	**M1**	Substitutes $y=mx$ with their nonzero m into the circle and solves for x, or finds the point of contact by any valid method.
	$w=\frac{12}{5}+\frac{9}{5}i$	**A1**	oe, such as $2.4+1.8i$.
	$\lvert w\rvert=\sqrt{2.4^2+1.8^2}$ $\arg w=\arctan\frac{1.8}{2.4}$	**M1**	Finds the modulus and argument of their w; the argument may be taken as their α.

Question 8 continued

Question	Scheme	Marks	Notes
	$w = 3.0(\cos 0.64 + i \sin 0.64)$	**A1**	In the form $r(\cos\theta + i\sin\theta)$; accept $r = 3$ and $\theta = 37°$.
		[4]	
	Total for Question 8	**8**	

Question	Scheme	Marks	Notes
9(a)	$w-2+i=(x-2)+4i$ $\tan\frac{\pi}{6}=\frac{4}{x-2}\Rightarrow x=\ldots$	**M1**	Equates $\tan\frac{\pi}{6}$ to the ratio of imaginary to real part of $w-2+i$ with $\text{Im}(w)=3$, and solves. Condone one sign slip.
	$w=2+4\sqrt{3}+3i$	**A1**	oe, or awrt $8.93+3i$.
		[2]	
9(b)	$k=\lvert\text{“}2+4\sqrt{3}+3i\text{”}-(2+11i)\rvert$ $=\lvert 4\sqrt{3}-8i\rvert$	**M1**	Uses Pythagoras on the real and imaginary differences between their w from (a) and $2+11i$. Condone sign slips.
	$k^2=112$	**A1**	Or $k=4\sqrt{7}$ oe, awrt 10.6.
	$y+1=\frac{x-2}{\sqrt{3}}$	**B1**	oe. The restriction $x>2$ need not be stated.
	$(x-2)^2+\left(\frac{x-2}{\sqrt{3}}-12\right)^2=\text{“}112\text{”}$	**M1**	Substitutes their linear equation into a circle centre $(2,11)$ with their k^2, reaching an equation in one variable.
	$x^2-(4+6\sqrt{3})x+28+12\sqrt{3}=0$	**A1**	Any correct three-term quadratic in one variable, oe, such as $y^2-4y+3=0$.
	$x=2+4\sqrt{3}$ (gives P) or $x=2+2\sqrt{3}$ $\Rightarrow y=\ldots$	**dM1**	Solves their quadratic by any method, calculator allowed, rejects the root giving P and finds the other coordinate.
	$z=2+2\sqrt{3}+i$	**A1**	oe, or awrt $5.46+i$.
		[7]	
Total for Question 9		**9**	

Question	Scheme	Marks	Notes
10(a)			
	Shape	**B1**	Major arc of a circle with endpoints $2+i$ and $6+5i$, both shown excluded.
	Labelled point	**B1**	Arc on the side of chord AB below and to the right, away from the origin.
		[2]	
10(b)	Centre equidistant from A and B, so on the perpendicular bisector of AB: midpoint $(4,3)$, gradient -1 $\Rightarrow y-3=-(x-4) \Rightarrow y=7-x$	**E1**	Gives the equidistance reason and shows the perpendicular bisector of AB is $y=7-x$, by midpoint and gradient or any valid check.
		[1]	
10(c)	$\angle APB = 2\times\frac{\pi}{4}=\frac{\pi}{2}$ $\Rightarrow r^2+r^2=AB^2=32 \Rightarrow r=\ldots$	**M1**	Uses the angle at the centre with Pythagoras on chord AB, or another valid chord relation such as $AB=2r\sin\frac{\pi}{4}$. Condone a slip in AB^2.
	$r=4$	**A1**	
		[2]	
10(d)	Centre $(a,7-a)$: $(a-2)^2+(6-a)^2=16$ $\Rightarrow a^2-8a+12=0 \Rightarrow a=\ldots$	**M1**	Uses a centre on $y=7-x$ at distance their r from A or B and solves for it. Condone expansion slips.

Question 10 continued

Question	Scheme	Marks	Notes
	$a = 2$ or 6: $(2, 5)$ or $(6, 1)$ $10 + i$ on locus, $\sqrt{80} \neq 4$ from $(2, 5)$ Centre $(6, 1)$	**A1**	Rejects $(2, 5)$ with a reason, such as a test point on the locus or the arc lying below and to the right of AB.
		[2]	
	Total for Question 10	**7**	